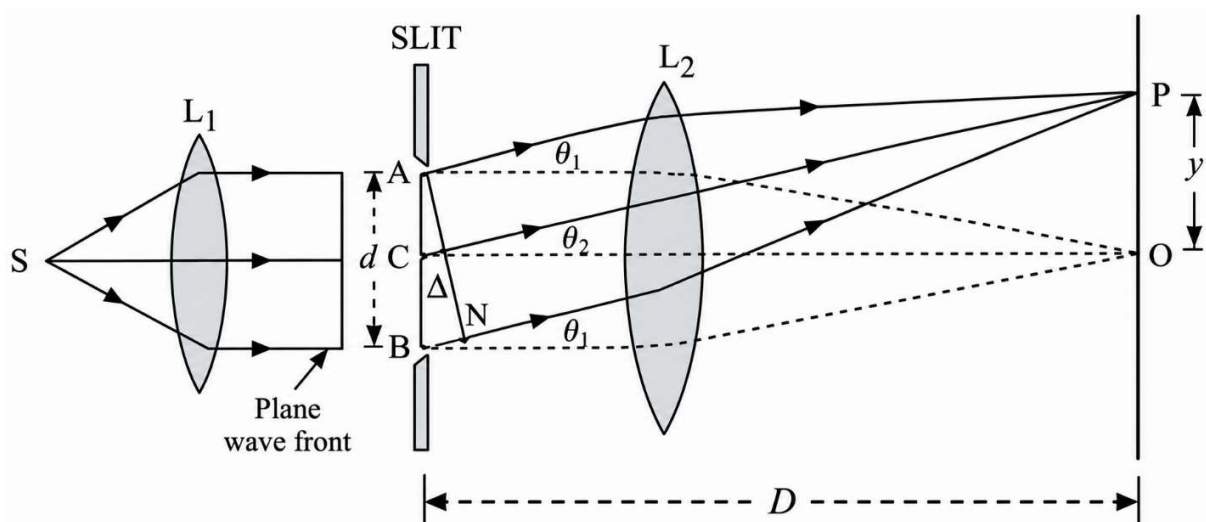


IILM UNIVERSITY, GREATER NOIDA		LABORATORY MANUAL
PRACTICAL EXPERIMENT INSTRUCTION SHEET		
EXPERIMENT TITLE: To study the intensity distribution due to diffraction from single slit and to determine the slit width (d).		
EXPERIMENT NO. 08	DEPT. : Physics	ISSUE DATE: 12 August, 2026
LABORATORY: 26SOSP101 Semiconductor and Quantum Physics Lab	SEMESTER: I, II	TOTAL PAGE: 04

AIM: To study the intensity distribution due to diffraction from single slit and to determine the slit width (d).

APPARATUS: Laser Source, Optical bench, screen with a rectangular slit, micro ammeter.

DIAGRAM:



FORMULA USED:

Light travels in a straight line. However, when light passes through a small hole, there is a certain amount of spreading of light. Similarly, when light passes by an obstacle, it appears to bend round the edges of the obstacle and enters its geometrical shadow. The phenomenon of bending of light around the corners of small obstacles or apertures and its consequent spreading into the regions of geometrical shadow is called diffraction of light.

Here we study the Fraunhofer class of diffraction in which light source is at infinite distance from obstacle, to ensure this source S of monochromatic light is placed at the focus of a convex lens L1.

A parallel beam of light and hence a plane wave front WW' gets incident on a narrow rectangular slit AB of width d.

Slit width 'd' is given by

$$d = \frac{2D\lambda}{\beta}$$

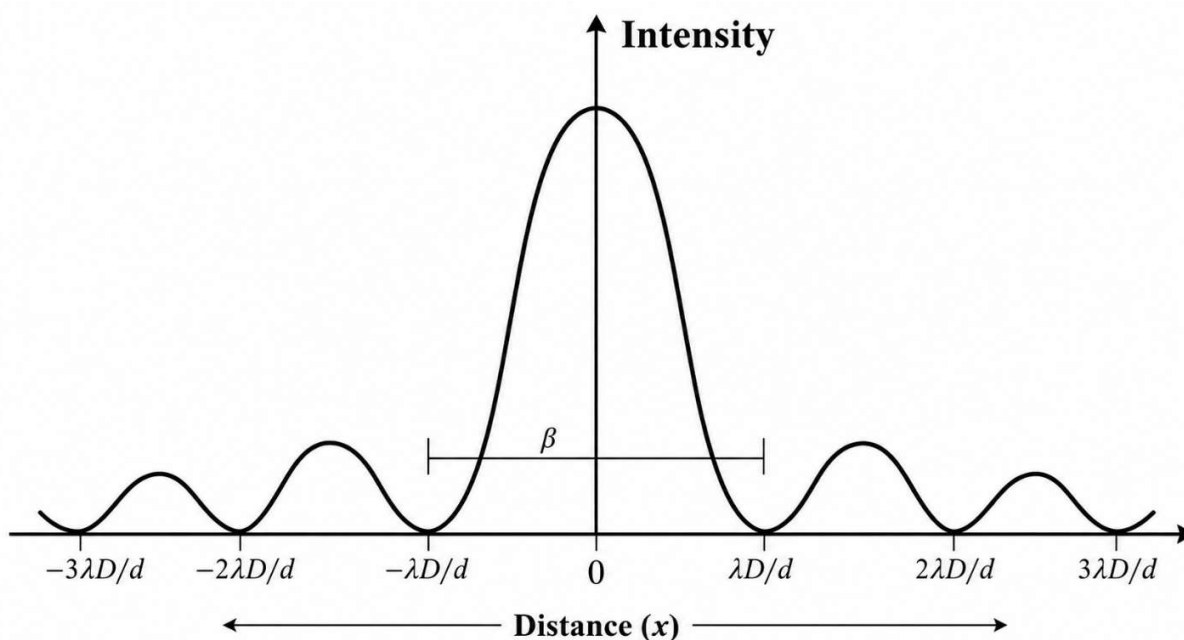
Where D = is the distance of screen from slit
 λ = Wavelength of laser (630 nm)
 β = width of central maxima

First minima on either side of Central maxima is given by condition for diffraction minima. viz,

$$d \sin \theta = m \lambda \quad \text{with } m = 1$$

PROCEDURE:

1. Switch on the laser source about 15 minutes before the experiment is due to start. This ensures that the intensity of light from the laser source is constant.
2. Allow the laser beam to fall on a slit formed in the screen provided. The photo detector is secured to a mount and is kept as far behind the slit as possible.
3. Try to observe the diffraction pattern by putting a screen (paper) in front of photo detector. After getting diffraction pattern on the screen (paper) remove it.
4. Adjust the position of photo detector somewhere around second minima & keep on recording the current with the position of photo detector moving it towards central maxima and continue in same direction till second minima on other side is obtained. The intensity distribution of the diffraction pattern is measured with the help of a photo detector connected to a microammeter.
5. Measure the distance between slit & photo detector.
6. Plot a graph (position of detector) vs (current). It will be of shape as shown in fig;



7. Now measure the distance (β) between two first minima on either side of central maxima from graph. This is the width of central maxima ' β '. Now calculate width of slit using formula.

OBSERVATIONS:

1. Wavelength of laser $\lambda=630$ nm
2. Position of Slit (a)= cm
3. Position of detector on bench (b) =.....

OBSERVATION TABLE:

Sr. No.	Distance between two first minimum on either side of central maxima (β in cm)	Distance between the detector & slit $D=(a-b)$ in cm	Slit width $d = \frac{2D\lambda}{\beta}$ (microns)	Mean d (microns)
1.				

2.				
3.				
4				

CALCULATIONS:

$$d = \frac{2D\lambda}{\beta}$$

RESULT:

1. The intensity distribution due to diffraction at a single slit was studied.
2. The width of central maxima ' β '=.....mm
3. The width of the single slit ismicrons.

PRECAUTIONS & SOURCES OF ERROR:

1. The laser beam should not penetrate into eyes as this may damage the eyes permanently.
2. The photo detector should be as away from the slit as possible.
3. The laser should be operated at a constant voltage 220V obtained from a stabilizer. This avoidsthe flickering of the laser beam.
4. Laser should be started at least 15 minutes before starting the experiment.

VIVA QUESTIONS:

1. Why is a monochromatic and coherent source like a laser necessary?
2. What would happen if you used polychromatic light instead of a laser?
3. What causes diffraction in the single slit experiment?
4. Which type of diffraction is used in this experiment?